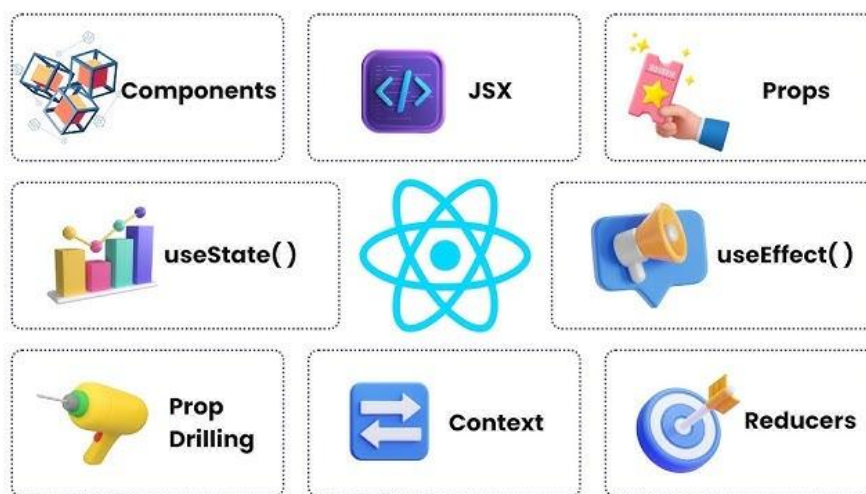


Key Concepts in React

What is React?

- React is a tool (JavaScript library) used to make websites and apps.
- It helps developers create fast and interactive user interfaces (UI).
- UI means what we see and click on a website.



Key Concepts in React

1. Components (Building Blocks)

- A **component** is a small part of a website.
- Websites are made by joining many components together.

Example:

```
function Welcome() {  
  return <h2>Hello Student!</h2>;  
}
```

Meaning:

This component shows **Hello Student!** on the screen.

2. JSX (Mixing HTML and JavaScript)

- JSX lets us write **HTML inside JavaScript**.

- It makes the code easy to read and design UI.

Example:

```
const element = <h1>Welcome to React</h1>;
```

Meaning:

We are writing HTML (h1) inside JavaScript.

3. Props (Passing Data to Components)

- **Props** are used to send data from one component to another.
- Think of it like **giving information to a friend**.

Example:

```
function Welcome(props) {  
  return <h2>Hello {props.name}</h2>;  
}
```

```
<Welcome name="Lucky" />
```

Output:

Hello Lucky

4. State (Remembering Data)

- **State** helps a component **remember information**.
- Example: counting likes or clicks.

Example:

```
import { useState } from "react";
```

```
function Counter() {  
  const [count, setCount] = useState(0);  
  return <h2>Count: {count}</h2>;  
}
```

Meaning:

React remembers the value of **count**.

5. Events and Conditional Rendering

- **Events** happen when a user clicks or types.
- **Conditional rendering** means showing something only when needed.

Example:

```
function Message() {  
  const isLoggedIn = true;  
  return <h2>{isLoggedIn ? "Welcome Back!" : "Please Login"}</h2>;  
}
```

Meaning:

If the user is logged in, show **Welcome Back!**

6. Lists and Keys

- React can display **many items in a list**.
- Each item needs a **key** so React can track it.

Example:

```
const fruits = ["Apple", "Mango", "Banana"];
```

```
function FruitList() {  
  return (  
    <ul>  
      {fruits.map((fruit, index) => (  
        <li key={index}>{fruit}</li>  
      ))}  
    </ul>  
  );  
}
```

Meaning:

React shows a list of fruits.

7. Hooks (Making React Smarter)

- **Hooks** help React add extra features like memory and updates.
- The most common hook is **useState**.

Example:

```
import { useState } from "react";
```

```
function LikeButton() {  
  const [likes, setLikes] = useState(0);  
  
  return (  
    <button onClick={() => setLikes(likes + 1)}>  
      Likes {likes}  
    </button>  
  );  
}
```

Meaning:

Every time you click, the **likes number increases**.

Quick Summary

- **Components** → Small parts of a website
- **JSX** → HTML inside JavaScript
- **Props** → Passing data
- **State** → Remembering data
- **Events** → User actions like clicks
- **Lists & Keys** → Showing many items
- **Hooks** → Extra power for React